

IRFAN AHAMED MELEKKANDY PUTHALATH

Newark, NJ — (862)423-3940 — irfan.aham1999@gmail.com

[LinkedIn](#) — [GitHub](#) — [Portfolio](#)

PROFESSIONAL SUMMARY

Software Development Engineer with **2.5 years** building scalable, distributed systems in **Java 17 and Python on AWS**. Proven record of high-availability microservices (**28k req/s at <30 ms p95**) and strong object-oriented design skills. Thrives in agile teams delivering secure code for **millions of users** and passionate about continuous improvement.

EXPERIENCE

CGI Groups Inc. Bengaluru, India

Software Engineer (Backend Systems) Sep 2022 – Jul 2024

Modernized legacy payments platform:

Decomposed Java EE monolith into

12 Spring Boot/

Vert.x

microservices

on AWS EKS, achieving

28k req/s at

<

30

ms

p95 latency

with zero-downtime deployments.

Reduced AWS spend

\$

1.8M annually:

Implemented EC2 right-sizing and spot fleet strategies, cutting costs while maintaining system performance and reliability.

Engineered exactly-once Kafka streams (99.99% availability):

Built Kafka stream processing pipelines for

1M+ daily transactions

, ensuring mission-critical system reliability.

Mentored 3 junior engineers:

Conducted code reviews and system design sessions, improving team communication and SQL tuning capabilities.

Championed secure coding practices:

Led quarterly threat modeling sessions, preventing critical CVEs across

8 production releases

.

Vishwas Coffee Coorg, India

Software Engineer (Full Stack) Sep 2021 – Aug 2022

End-to-end ownership of supply-chain digitization:

Delivered Python/

Flask+MySQL

solution, increasing inventory accuracy by

25%

and streamlining operations.

Barcode-based stock tracking and Angular dashboard:

Built automated tracking system and Angular dashboard, reducing fulfillment time by

30%

and boosting repeat orders

15%

.

Drove CI/CD with

PyTest

+ Jest and Docker Compose:

Implemented automated testing and deployment pipeline, cutting production defects by

40%

.

TECHNICAL PROJECTS

GridCast (Energy Demand Forecasting API) *Mar 2025*

Deployed Python

FastAPI

service on AWS Lambda:

Built energy demand prediction API

15% more accurate

than baseline, serving results in

10

ms

via REST endpoint.

Designed CI/CD workflow with GitHub Actions and Terraform Cloud:

Achieved

98% pipeline reliability

across

150+ automated runs

with consistent deployments.

Apna-Pharmacy (Cloud-Native Rx Platform) *Aug 2022*

Architected Spring Boot + Kafka + Kubernetes microservices:

Built cloud-native platform processing

3k+ prescriptions/day

, reducing latency

20%

through horizontal pod autoscaling.

Optimized PostgreSQL with advanced indexing:

Implemented indexing strategies, achieving

5

×

faster query times

and improved system responsiveness.

EDUCATION

Rutgers University Newark, NJ

Master of Science in Information Technology *Sep 2024 – May 2025 (Expected)*

Focus: Artificial Intelligence & Machine Learning **GPA: 3.84**

Key Courses: Distributed Systems, Cloud Architecture, Advanced Databases

RV College of Engineering Bengaluru, India

Bachelor of Engineering in Electronics & Instrumentation *Aug 2017 – Aug 2021*

TECHNICAL SKILLS

Languages: Java (17/21), Python, SQL, JavaScript/TypeScript, Bash, C++, Go (basic)

Frameworks & Libraries: Spring Boot, Vert.x, Quarkus, Flask, React, Angular, gRPC, GraphQL

Cloud & DevOps: AWS (EC2, S3, EKS, RDS, Lambda, Kinesis), Docker, Kubernetes, Terraform, GitHub Actions,

Argo CD, Jenkins, Git

Databases: PostgreSQL, MySQL, Redis, MongoDB, DynamoDB, Elasticsearch

Architecture: Microservices, Distributed Systems, REST APIs, High Availability, Scalability, Object-Oriented Design

Tools: Kafka, Spark (Pyspark), Prometheus, Grafana, Jaeger, OpenTelemetry

Process & Methodologies: Agile/Scrum, Test-Driven Development, CI/CD, SLO Monitoring, Code Reviews, Technical

Mentoring, Problem Solving, Customer Obsession, Teamwork

Certifications: AWS Certified Developer – Associate, Docker Essentials (Udemy)